

Place value



1 Write down the place value represented by the digit 7 in the following numbers:

a 48.07

b 0.487

c 47.008

d 40.708

2 How do you say the following numbers?

a 59.412

b 6.026

3 Write the number 'sixty five and six hundred and eighty nine thousandths' as a decimal.

4 Write the following numbers in decimal form:

a Four tenths

b Seventy hundredths

c Seventy six hundredths

d Two and three tenths

e Six and seven hundredths

f Twelve and twenty-two thousandths

5 Write the following fractions as decimals:

a $\frac{7}{10}$

b $\frac{94}{100}$

c $\frac{61}{1000}$

d $\frac{689}{1000}$

e $\frac{2}{10}$

f $\frac{9}{100}$

g $\frac{7}{1000}$

h $\frac{63}{1000}$

6 For each of the following, fill in the boxes with the missing numbers:

a $8.43 = 8 + \square + 0.03$

b $29.65 = 20 + 9 + \square + 0.05$

c $8.52 = 8 + \square$

d $31.305 = 30 + 1 + \square + \square$

Expanded form

7 Convert the following numbers from expanded form to decimal form:

a $2 \times 10 + 5 \times 1 + 7 \times \frac{1}{10} + 3 \times \frac{1}{100} + 2 \times \frac{1}{1000}$

b $1 \times 10 + 0 \times 1 + 2 \times \frac{1}{10} + 0 \times \frac{1}{100} + 1 \times \frac{1}{1000}$

c $3 \times 100 + 0 \times 10 + 3 \times 1 + 0 \times \frac{1}{10} + 1 \times \frac{1}{100} + 4 \times \frac{1}{1000}$

d $4 \times 1 + 0 \times \frac{1}{10} + 0 \times \frac{1}{100} + 1 \times \frac{1}{1000}$

8 Consider the number read as "34 and 695 hundredths":



a Write the number in expanded form.

b Write the number in decimal form.

9 Write the following numbers in decimal form:

a $7 + \frac{4}{10} + \frac{2}{100}$

b $75 + \frac{8}{10} + \frac{2}{100}$

c $500 + 2 + \frac{8}{10} + \frac{3}{1000}$

d $12 + \frac{2}{100} + \frac{3}{10000}$

Compare decimals

10 For each of the following, choose the larger decimal number:

a 0.71 or 0.7

b 0.39 or 0.4

c 2.305 or 2.503

d 4.9999 or 4.8

11 Which is the smaller decimal: 0.76 or 0.67?

12 Write the following numbers in ascending order:

a 0.24, 0.204, 2.4, 0.024, 0.0024

b 3.68, 3.86, 3.068, 3.681, 3.1

13 Write the following numbers in descending order:

a 0.54, 0.504, 0.5, 0.054, 0.0054

b 7.28, 7.82, 7.078, 7.168, 7.8

14 For each of the following, insert the symbol $>$ or $<$ to make the expression true:

a $0.71 \square 0.7$

b $0.39 \square 0.389$

c $2.305 \square 2.309$

d $4.9998 \square 4.9999$

e $0.001 \square 0.0009$

f $2.189 \square 2.19$

g $0.035 \square 0.0043$

h $0.99999 \square 1$

Multiply and divide by powers of 10

15 Consider the product 0.973×100 .

a State in which direction the decimal point must be moved to calculate the answer.

b State the number of places the decimal point must be moved. Explain your answer.

c Hence calculate the product.



 **16** Consider the fraction $\frac{452}{1000}$.

- a** Write the fraction as a quotient using a ' $\div$ ' sign.
- b** 452 is equivalent to 452.0. State in which direction the decimal point in 452.0 must be moved to calculate the answer to part (a).
- c** State the number of places the decimal point must be moved. Explain your answer.
- d** Hence write the quotient as a decimal number.

 **17** Evaluate, writing your answer in decimal form:

- | | | | |
|------------------------------|-------------------------------|------------------------------|------------------------------|
| a 0.7×10 | b 0.0098×10 | c 0.02×10 | d 0.65×10 |
| e 0.093×10 | f 0.3784×10 | g 5.2×10 | h 0.003×10 |
| i 0.35×100 | j 9.3×100 | k 6.74×100 | l 59.43×100 |
| m 0.035×100 | n 0.007×100 | o 0.4×100 | p 0.0046×100 |
| q 0.204×100 | r 0.6489×1000 | s 0.04×1000 | t 5.761×1000 |
| u 0.008×1000 | v 0.0043×1000 | w 0.408×1000 | x 0.934×1000 |

 **18** Evaluate, writing your answer in decimal form:

- | | | | |
|---------------------------|----------------------------|-----------------------------|----------------------------|
| a $0.3 \div 10$ | b $4.707 \div 10$ | c $34 \div 10$ | d $0.57 \div 10$ |
| e $0.832 \div 10$ | f $3.9 \div 10$ | g $6.75 \div 10$ | h $7.651 \div 10$ |
| i $95.71 \div 10$ | j $79.66 \div 100$ | k $8.4 \div 100$ | l $0.2 \div 100$ |
| m $56 \div 100$ | n $423 \div 1000$ | o $953.5 \div 100$ | p $423.3 \div 1000$ |
| q $5.6 \div 100$ | r $0.42 \div 100$ | s $7583.5 \div 100$ | t $2.2 \div 1000$ |
| u $\frac{52}{100}$ | v $\frac{104}{100}$ | w $\frac{203}{1000}$ | x $\frac{2}{100}$ |